Equivalent characterisations of planar orthogonal polynomials (Theorem 2.1)

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned source file sources/sigma-2023-020-planar-orthogonal-polynomials-as-type-i-mult/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span is the author Theorem environment with its own label; the proof is delivered as two spans - the main proof (source lines 413-545, closing 'completes the proof of Theorem') and the author's alternative proof of (a) implies (c) (source lines 554-604) with its introducing equation (source lines 548-552) as a context span.

Editorial note. The proof takes Q(z) = z^k in the fundamental identity (eq:PQscalar); the identity's carrier Proposition prop:Contour1 (printed 3.1) is delivered as a context span placed after the statement, so it prints locally renumbered and the statement keeps 2.1.

Editorial note. Disclosure: the source states the three operator properties W1-W3 (source lines 448-475) without proof, and step (c)-(d) invokes the uniqueness theorem for homogeneous constant-coefficient linear ODEs (source lines 513-514); both are named standard facts the argument reduces to.

Editorial note. Printed number anchored to the published PDF (SIGMA 19 (2023) 020, read from pdftotext of sigma23-020.pdf). The pinned tex has seven sections and no appendix and its section structure matches the published PDF one-to-one, so the unit 010 pinned-versus-published realignment does not trigger.

Editorial note. Equation and theorem numbers inside the card are locally renumbered; every cross-reference is a symbolic \eqref/\ref and resolves inside the card.

Editorial scope. Single environment card cut from the SIGMA article doi:10.3842/SIGMA.2023.020 (sigma-2023-020-planar-orthogonal-polynomials-as-type-i-mult); statement and proof are the author's own text at source lines 215-244 and 413-545.

Allowed standard prerequisites (editorial).

- Fundamental identity (Proposition 3.1). For polynomial W with positive integer exponents the W-scalar product equals a contour integral of P(z)W(z) against a ray integral of W*(u)Q*(u) e^{-uz} (eq:PQscalar, delivered in context).. Load-bearing: step (a)-(b) takes Q(z) = z^k in this identity and reads off eq:Pzk.

- Scalar product and orthogonality system of P_n. The scalar product of the modified Gaussian measure and the defining vanishing-moment system of P_n (eq:fgscalar, def:Pn, delivered in context).. Load-bearing: the equivalence is stated for the monic polynomial P_n of this system; the alternative proof cites def:Pn directly.
- Operator properties W1, W2, W3. Stated inside the main proof span (source lines 448-475): kernel description eq:Wproperty1, degree behaviour eq:Wproperty2/eq:Wproperty2alt, vanishing-order reduction eq:Wproperty3.. Load-bearing: the (c)-(d) equivalences apply them explicitly; the source states them as basic properties of the product-form operator and proves none of them.
- Uniqueness theorem for constant-coefficient linear ODEs. A solution of a homogeneous constant-coefficient linear ODE of order c with c vanishing initial conditions is identically zero.. Load-bearing: step (c)-(d) concludes $Q = 0$ from the c vanishing initial conditions (source lines 513-514); standard named theorem, stated without proof in the source.

The polynomials in question are orthogonal with respect to a modified Gaussian measure,

$$\mu_W(\mathrm{d}z) = \frac{1}{\pi}|W(z)|^2 \mathrm{e}^{-|z|^2}\,\mathrm{Leb}(\mathrm{d}z), \tag{2.1}$$

where Leb denotes the Lebesgue measure on $\mathbb{C}$ (identified with $\mathbb{R}^2$) and the weight W reads

$$W(z) = \prod_{j=1}^{p}(z-a_j)^{c_j}, \qquad z \in \mathbb{C}, \tag{2.2}$$

where $p \in \mathbb{N}$, the c_j are positive real numbers, and the a_j are distinct complex numbers.

Denote the scalar product corresponding to (2.1) by

$$\langle f, g\rangle_W = \int_{\mathbb{C}} f(z)\overline{g(z)}\,\mu_W(\mathrm{d}z). \tag{2.3}$$

Then, the n-th degree monic orthogonal polynomial P_n with respect to μ_W can be uniquely recovered by solving a linear system of equations for its coefficients,

$$\int_{\mathbb{C}} P_n(z)\overline{z}^k\,\mu_W(\mathrm{d}z) = 0, \qquad k = 0, \dots, n-1. \tag{2.4}$$

Our main result is Theorem 2.1 below. It gives a number of properties that are equivalent to the planar orthogonality corresponding to (2.1)–(2.4) in the case the c_j are positive integers. Below, we use the conjugate W^* of W defined by

$$W^*(z) = \overline{W(\bar{z})} = \prod_{j=1}^{p}(z-\overline{a_j})^{c_j}, \qquad z \in \mathbb{C}. \tag{2.5}$$

We also use $\mathcal{D}_z$ to denote the derivative operator with respect to z, and then $W^*(\mathcal{D}_z)$ is the differential operator

$$W^*(\mathcal{D}_z) = \prod_{j=1}^{p}(\mathcal{D}_z-\overline{a_j})^{c_j}. \tag{2.6}$$

Theorem 2.1. *Let W be given by* (2.2) *where all c_j are positive integers* $\big($*so that W is a polynomial of degree $c = \sum_{j=1}^{p} c_j$*$\big)$, and let the a_j be distinct complex numbers. Then, the following properties are equivalent for a monic polynomial P_n of degree n,*

(*a*) *P_n is the planar orthogonal polynomial on $\mathbb{C}$ with weight* (2.1)*.*

(*b*) *P_n satisfies*

$$\frac{1}{2\pi \mathrm{i}} \oint_{\gamma} P_n(z) W(z) \phi_k(z)\, \mathrm{d}z = 0, \qquad k = 0, 1, \ldots, n-1, \tag{2.7}$$

where γ is a closed contour around the origin and

$$\phi_k(z) = \int_0^{\bar{z}\times\infty} W^*(u) u^k \mathrm{e}^{-uz}\, \mathrm{d}u, \tag{2.8}$$

where the path of integration in (2.8) *goes from 0 to ∞ along the ray $\arg u = \arg \bar{z}$.*

(*c*) *One has*

$$W^*(\mathcal{D}_z)\,[P_n(z)W(z)] = O(z^n) \tag{2.9}$$

as $z \to 0$.

(*d*) *There exist polynomials Q_j of $\deg Q_j \le c_j - 1$, for $j = 1, \ldots, p$, such that*

$$P_n(z)W(z) + \sum_{j=1}^{p} Q_j(z) \mathrm{e}^{\overline{a_j} z} = O\big(z^{n+c}\big) \tag{2.10}$$

as $z \to 0$.

Proposition 2.2. *Let P and Q be polynomials and suppose that the c_j in* (2.2) *are positive integers. Then, the fundamental identity holds,*

$$\begin{aligned} \langle P, Q\rangle_W &= \frac{1}{\pi} \int_{\mathbb{C}} P(z)\overline{Q(z)} |W(z)|^2 \mathrm{e}^{-|z|^2}\, \mathrm{Leb}(\mathrm{d}z) \\ &= \frac{1}{2\pi \mathrm{i}} \oint_{\gamma} P(z)W(z) \int_0^{\bar{z}\times\infty} W^*(u) Q^*(u) \mathrm{e}^{-uz}\, \mathrm{d}u\, \mathrm{d}z, \end{aligned} \tag{2.11}$$

where γ is a simple closed contour that goes around the origin once in the counterclockwise direction and the path for the u integral goes from 0 to ∞ along the ray $\arg u = \arg \bar{z}$.

Proof. $(a) \Leftrightarrow (b)$: Taking $Q(z) = z^k$ in the identity (2.11) and recalling the definition (2.8) of ϕ_k, we obtain

$$\big\langle P, z^k \big\rangle_W = \frac{1}{2\pi \mathrm{i}} \oint_{\gamma} P(z) W(z) \phi_k(z)\, \mathrm{d}z. \tag{2.12}$$

This identity (2.12) shows that (a) and (b) in Theorem 2.1 are equivalent.
$(b) \Leftrightarrow (c)$: From basic properties of the Laplace transform, we get

$$\int_0^{\bar{z}\times\infty} W^*(u) u^k \mathrm{e}^{-uz}\, \mathrm{d}u = W^*(-\mathcal{D}_z)(-\mathcal{D}_z)^k \left[\frac{1}{z}\right] = W^*(-\mathcal{D}_z)\left[\frac{k!}{z^{k+1}}\right],$$

Thus, recalling (2.8), we can write

$$\frac{1}{2\pi\mathrm{i}}\oint_\gamma P_n(z)W(z)\phi_k(z)\,\mathrm{d}z = \frac{k!}{2\pi\mathrm{i}}\oint_\gamma P_n(z)W(z)W^*(-\mathcal{D}_z)\left[\frac{1}{z^{k+1}}\right]\mathrm{d}z$$

for any polynomial P_n. Integrating by parts, we find

$$\frac{1}{2\pi\mathrm{i}}\oint_\gamma P_n(z)W(z)\phi_k(z)\mathrm{d}z = \frac{k!}{2\pi\mathrm{i}}\oint_\gamma \frac{W^*(\mathcal{D}_z)[P_n(z)W(z)]}{z^{k+1}}\,\mathrm{d}z,$$

as there are no boundary terms on the closed contour γ and $W^*(\mathcal{D}_z)$ is the adjoint of $W^*(-\mathcal{D}_z)$. Thus, P_n satisfies (2.7) if and only if

$$\frac{1}{2\pi\mathrm{i}}\oint_\gamma \frac{W^*(\mathcal{D}_z)[P_n(z)W(z)]}{z^{k+1}}\,\mathrm{d}z = 0, \qquad k = 0,\dots,n-1. \tag{2.13}$$

The identities (2.13) mean that the coefficient before z^k of the polynomial $W^*(\mathcal{D}_z)[P_n(z)W(z)]$ vanishes for $k = 0,\dots,n-1$, and we conclude that items (b) and (c) are equivalent.

In the proof that (c) and (d) are equivalent, we are going to use three basic properties of the operator $W^*(\mathcal{D}_z)$ from (2.6). These properties are stated separately for the ease of reference.

Property W_1. $W^*(\mathcal{D}_z)$ is a linear differential operator with kernel

$$\ker W^*(\mathcal{D}_z) = \left\{ \sum_{j=1}^{p} Q_j(z)\mathrm{e}^{\overline{a_j}z} \;\middle|\; \deg Q_j \le c_j - 1 \text{ for } j = 1,\dots,p \right\}. \tag{2.14}$$

The kernel is a vector space over $\mathbb{C}$ and its dimension is $c = \sum_{j=1}^{p} c_j$.

Property W_2. If P is a polynomial then $W^*(\mathcal{D}_z)[P(z)]$ is a polynomial and

$$\deg W^*(\mathcal{D}_z)[P(z)] = \deg P(z), \tag{2.15}$$

provided all $a_j \neq 0$. If $a_1 = 0$ and $\deg P(z) \ge c_1$, then

$$\deg W^*(\mathcal{D}_z)[P(z)] = \deg P(z) - c_1. \tag{2.16}$$

Property W_3. $W^*(\mathcal{D}_z)$ reduces the order of vanishing at $z = 0$ by $c = \deg W$, provided the order of vanishing is greater than or equal to c. Namely, for every non-negative integer n, and analytic function F with $F(z) = O(z^{n+c})$ as $z \to 0$, we have

$$W^*(\mathcal{D}_z)[F(z)] = O(z^n) \tag{2.17}$$

as $z \to 0$.

$(d) \Rightarrow (c)$: Suppose P_n satisfies (2.10). Then, by applying $W^*(\mathcal{D}_z)$ and using Properties W_1 and W_3, we obtain

$$W^*(\mathcal{D}_z)[P_n(z)W(z)] = W^*(\mathcal{D}_z)\big[O\big(z^{n+c}\big)\big] = O(z^n), \tag{2.18}$$

which is (2.9). Thus, (d) implies (c).

$(c) \Rightarrow (d)$: In the proof, we assume without loss of generality that $a_1 = 0$ and $c_1 \ge 0$. This includes the case that W has no zero at the origin since one can always set $c_1 = 0$.

Consider the linear mapping $\pi\colon \ker(W^*(\mathcal{D}_z)) \to \mathbb{C}^c$ given by

$$\ker(W^*(\mathcal{D}_z)) \ni Q(z) = \sum_{j=0}^{\infty} q_j z^j \overset{\pi}{\longmapsto} (q_0, \dots, q_{c_1-1}, q_{n+c_1}, \dots, q_{n+c-1}) \in \mathbb{C}^c. \tag{2.19}$$

We claim that π is injective, which will imply that π is surjective since it is a linear mapping between vector spaces of the same dimension c, as follows from (2.19) and Property W_1. Suppose $Q \in \ker(W^*(\mathcal{D}_z))$ with $\pi(Q) = (0, \dots, 0) \in \mathbb{C}^c$. Then,

$$Q(z) = \sum_{j=c_1}^{n+c_1-1} q_j z^j + O\big(z^{n+c}\big) \tag{2.20}$$

as $z \to 0$. From (2.17) and $W^*(\mathcal{D}_z)[Q(z)] = 0$, it then follows that

$$W^*(\mathcal{D}_z)\left[\sum_{j=c_1}^{n+c_1-1} q_j z^j\right] = O(z^n) \tag{2.21}$$

as $z \to 0$. Because of (2.15)–(2.16), we have that the left-hand side of (2.21) is a polynomial of degree $\leq n-1$, and due to (2.21) it has a zero at $z = 0$ of order at least n. Hence, (2.21) vanishes identically, and thus $\sum_{j=c_1}^{n+c_1-1} q_j z^j$ belongs to the kernel of $W^*(\mathcal{D}_z)$. By Property W_1, the kernel contains polynomials up to degree $c_1 - 1$ but no polynomials of higher degrees. Hence, $q_j = 0$ for $j = c_1, \dots, n + c_1 - 1$. Consequently, by (2.20) we have that $Q(z) = O(z^{n+c})$ as $z \to 0$. In particular, $Q(0) = Q'(0) = \cdots = Q^{(c-1)}(0) = 0$. Thus $Q \in \ker(W^*(\mathcal{D}_z))$ is a solution of a homogeneous constant coefficient linear ODE of order c with c vanishing initial conditions. The uniqueness theorem for such ODEs then yields $Q \equiv 0$, which justifies the claim.

Now, assume that P_n satisfies (2.9), and write

$$P_n(z)W(z) = \sum_{j=0}^{n+c} p_j z^j.$$

Since π is surjective, there is $Q \in \ker(W^*(\mathcal{D}_z))$,

$$Q(z) = \sum_{j=0}^{\infty} q_j z^j,$$

such that $\pi(Q) = (-p_0, \dots, -p_{c_1-1}, -p_{n+c_1}, \dots, -p_{n+c-1})$. Then,

$$P_n(z)W(z) + Q(z) = \sum_{j=c_1}^{n+c_1-1} (p_j + q_j) z^j + O\big(z^{n+c}\big). \tag{2.22}$$

Applying $W^*(\mathcal{D}_z)$ to (2.22) and using (2.9), $W^*(\mathcal{D}_z)[Q(z)] = 0$, and (2.17), we obtain

$$W^*(\mathcal{D}_z)\left[\sum_{j=c_1}^{n+c_1-1} (p_j + q_j) z^j\right] = O(z^n) \tag{2.23}$$

as $z \to 0$. We find ourselves in the situation similar to that while proving the injectivity of π. It followed from (2.21) that $q_j = 0$ for $j = c_1, \dots, n + c_1 - 1$. In the exactly same way, now it follows from (2.23) that $p_j + q_j = 0$ for $j = c_1, \dots, n + c_1 - 1$.

Due to (2.22), we obtain

$$P_n(z)W(z) + Q(z) = O\big(z^{n+c}\big),$$

which is exactly (2.10) because $Q \in \ker(W^*(\mathcal{D}_z))$ must be of the form given in (2.14). This shows that (c) implies (d) and completes the proof of Theorem 2.1. □

We present an alternative proof of $(a) \Rightarrow (c)$ by making use of the Fourier transform of the measure μ_W in (2.1),

$$\widehat{\mu}_W(\zeta, \overline{\zeta}) = \int_{\mathbb{C}} \mathrm{e}^{\overline{\zeta} z + \zeta \overline{z}} \mu_W(\mathrm{d}z). \tag{2.24}$$

Alternative proof. $(a) \Rightarrow (c)$: Apply the Wirtinger derivatives

$$\mathcal{D}_\zeta = \frac{1}{2}\Big(\frac{\partial}{\partial u} - \mathrm{i}\frac{\partial}{\partial v}\Big), \qquad \mathcal{D}_{\overline{\zeta}} = \frac{1}{2}\Big(\frac{\partial}{\partial u} + \mathrm{i}\frac{\partial}{\partial v}\Big),$$

where $\zeta = u + \mathrm{i}v$, to (2.24), and observe that

$$\mathcal{D}_{\overline{\zeta}}^j \mathcal{D}_\zeta^k \big[\widehat{\mu}_W(\zeta, \overline{\zeta})\big] = \int_{\mathbb{C}} z^j \overline{z}^k \mathrm{e}^{\overline{\zeta} z + \zeta \overline{z}} \mu_W(\mathrm{d}z). \tag{2.25}$$

Due to linearity, (2.25) leads to

$$P_n(\mathcal{D}_{\overline{\zeta}}) \mathcal{D}_\zeta^k \big[\widehat{\mu}_W(\zeta, \overline{\zeta})\big] = \int_{\mathbb{C}} P_n(z) \overline{z}^k \mathrm{e}^{\overline{\zeta} z + \zeta \overline{z}} \mu_W(\mathrm{d}z).$$

Assuming that P_n is the planar orthogonal polynomial satisfiying (2.4), we have

$$P_n(\mathcal{D}_{\overline{\zeta}}) \mathcal{D}_\zeta^k \big[\widehat{\mu}_W(\zeta, \overline{\zeta})\big]\big|_{\zeta=0, \overline{\zeta}=0} = 0, \qquad k = 0, \dots, n-1, \tag{2.26}$$

where we view ζ and $\overline{\zeta}$ as two independent variables. The latter is permissible as long as the Fourier transform $\widehat{\mu}_W(\cdot, \cdot)$, as a function of two complex variables, extends analytically to the neighborhood of $\big\{(\xi_1, \xi_2) \in \mathbb{C} \times \mathbb{C} \mid \xi_2 = \overline{\xi_1}\big\}$. This is the case we encounter below.

For μ_W from (2.1) with polynomial W, the Fourier transform can be computed as follows,

$$\begin{aligned}
\widehat{\mu}_W(\zeta, \bar{\zeta}) &= \frac{1}{\pi} \int_{\mathbb{C}} \mathrm{e}^{\overline{\zeta} z + \zeta \overline{z}} |W(z)|^2 \mathrm{e}^{-|z|^2} \operatorname{Leb}(\mathrm{d}z) \\
&= \frac{1}{\pi} W(\mathcal{D}_{\overline{\zeta}}) W^*(\mathcal{D}_\zeta) \int_{\mathbb{R}^2} \mathrm{e}^{2(ux+vy)} \mathrm{e}^{-x^2-y^2} \, \mathrm{d}x \, \mathrm{d}y \\
&= W(\mathcal{D}_{\overline{\zeta}}) W^*(\mathcal{D}_\zeta) \big[\mathrm{e}^{\zeta \overline{\zeta}}\big],
\end{aligned} \tag{2.27}$$

where $\zeta = u + \mathrm{i}v$, $z = x + \mathrm{i}y$.

Inserting (2.27) into (2.26), and changing the order of the differential operators, we obtain

$$\mathcal{D}_\zeta^k W^*(\mathcal{D}_\zeta)(P_n W)(\mathcal{D}_{\overline{\zeta}})\big[\mathrm{e}^{\zeta\overline{\zeta}}\big]\big|_{\zeta=0,\overline{\zeta}=0} = 0, \qquad k = 0, \dots, n-1. \tag{2.28}$$

Since

$$(P_n W)(\mathcal{D}_{\overline{\zeta}})\big[\mathrm{e}^{\zeta\overline{\zeta}}\big]\big|_{\overline{\zeta}=0} = P_n(\zeta)W(\zeta),$$

we see that (2.28) implies

$$\mathcal{D}_\zeta^k W^*(\mathcal{D}_\zeta)[P_n(\zeta)W(\zeta)]\big|_{\zeta=0} = 0, \qquad k = 0, \dots, n-1,$$

which is equivalent to (2.9). □